Probabilities: Modern and Old

Classical definition:

$$P(E) = \frac{\text{nr. of favorable outcomes}}{\text{total nr. of outcomes}}$$

$\hookrightarrow$ event

Problem: Roll a fair die
What's the probability to obtain an even nr.?

$$P(E) = \frac{3}{6} = \frac{1}{2}$$

$\dashrightarrow$ 2, 4, 6

Here the classical definition works well !!!

Problem: Roll a rigged die with 2 instead of 5.
What's the probability to obtain an even nr.?

$$P(E) \neq \frac{3}{5} = 60\,\%$$

2, 2, 4, 6
they count as 1

The probability is not the value 3/5 given by the clasical definition of probability---- That's the idea !

the correct probability is $P(E) = 4/6$

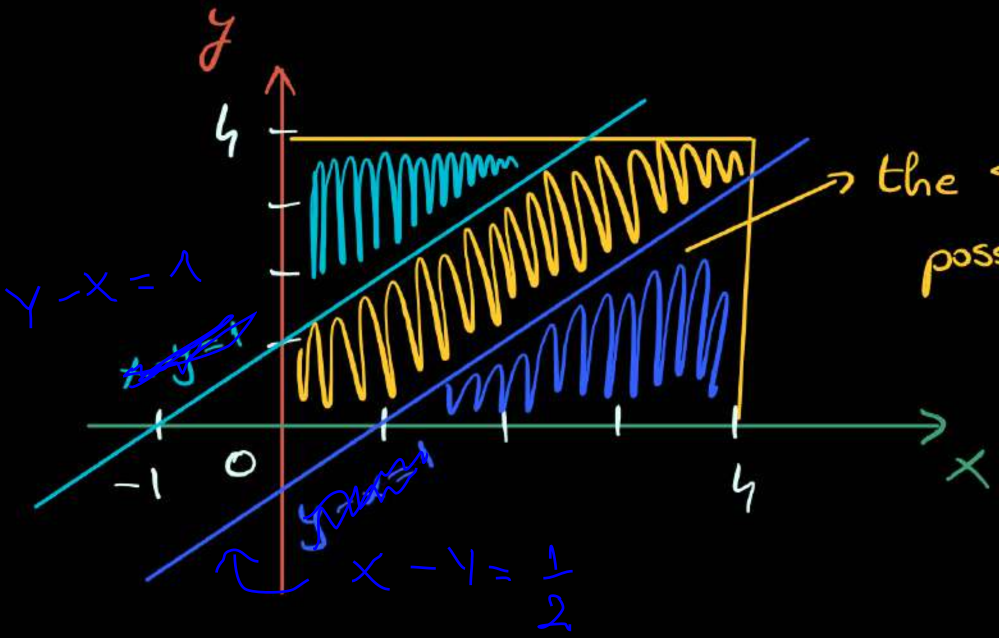
Problem of meeting:

A man and a woman decide to meet at a restaurant after 8 o'cloch. The restaurant closes at 12 o'cloch. The man waits for no more than 1 hour and the woman for no more than 30 minutes. (the difference, my god :))

What's the probability that they will meet at that restaurant?

Denote with x - the time when the man arrives
y - the time when the woman arrives

(x, y) - is a possible outcome



the set of all possible outcomes

When do they meet?

if the man arrives first $\Rightarrow y - x \leq \cancel{1}\ \frac{1}{2}$ (30 minutes)

if the woman arrives first $\Rightarrow x - y \leq \cancel{1}\ \frac{1}{2}$ (30 minutes/ ~~1 hour~~)

$\frac{x}{a} + \frac{y}{b} = 1 \Rightarrow$ it intersects the axes in $(a, 0)$ and $(0, b)$

$$\Rightarrow P(E) = \frac{A_{\square} - A_{\triangledown} - A_{\triangle}}{A_{\square}} \cong 0.33 = 33\%$$

$A_{\square}$ — Area of the ~~nr~~ of possible outcomes

Area of the region that contains the pairs (x,y) corresponding to the possible outcomes

!! in some real-life situations, the classical definition of probability fails.

=> a modern approach

Random experiment:

it's a process with 2 or more different outcomes, and with uncertainty as which will be observed.

Random experiment → roll a die

↓

Sample space = the set of all the possible outcomes of a random experiment

⇓

$S = \{1, 2, 3, 4, 5, 6\}$

↳ sample space when you roll a die

Elementary events = one particular possible outcome

Composite events = more than one particular outcome

roll a die

E = the event: we got number 1

$E = \{1\}$ → i. an elementary event

F = the event: we got an even number

$F = \{2, 4, 6\}$ → ii. composite event

Event = a subset of the sample space

The Axioms of Probability

A probability measure $P(\cdot)$ is a function that maps events from a sample space to a real number, such that:

(Axiom 1) $P(S) = 1$

(Axiom 2) for every event E, $P(E) \geq 0$

(Axiom 3) for each countable collection of mutually exclusive events $E_1, E_2, \dots \Rightarrow$

$$\Rightarrow P(E_1 \cup E_2 \cup \dots) = P(E_1) + P(E_2) + \dots$$

Possible Consequences of these Axioms

- $P(\emptyset) = 0$
- $P(E) \in [0, 1]$
- $P(\underbrace{E_1 \cup E_2}_{\text{either } E_1 \text{ or } E_2}) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$
- $P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_1 \cap E_3) - P(E_2 \cap E_3) + P(E_1 \cap E_2 \cap E_3)$
- $E_i \cap E_j = \emptyset, \forall i \neq j$

What happens when $E_1, E_2, \dots, E_n$ are not mutually exclusive?

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = \sum_{i=1}^{n} P(E_i) - \sum_{i<j} P(E_i \cap E_j)$$
$$+ \sum_{i<j<h} P(E_i \cap E_j \cap E_h) + \dots$$
$$+ (-1)^{p+1} \cdot \sum_{\cancel{i<j<\dots<p}} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_p}) \dots$$
$$i_1 < i_2 < \dots < i_p$$

- if $E_1 \subseteq E_2 \Rightarrow P(E_1) \leq P(E_2)$

The Boy-Girl Paradox:

$\boxed{A}$ Mister Smith has 2 children.
if at least one of them is a boy, what is the probability that the other one is a girl?

$\boxed{B}$ Mister Smith has 2 children.
if at least one of them is a boy born on Tuesday, what's the probability that the other one is a girl?

Let's think about situation $\boxed{B}$:
How do we build the sample space?

B_i - the event: the child is a boy and is born on the i-th day of the week
G_i - the child is a girl and is born on the i-th day of the week

$$S = \left\{ \begin{matrix} B_2B_1 \\ B_2B_2 \\ B_2B_3 \\ \vdots \\ B_2B_7 \end{matrix}, \quad \begin{matrix} B_1B_2 \\ \boxed{} \\ B_3B_2 \\ \vdots \\ B_7B_2 \end{matrix}, \quad \begin{matrix} B_2G_1 \\ B_2G_2 \\ B_2G_3 \\ \vdots \\ B_2G_7 \end{matrix}, \quad \begin{matrix} G_1B_2 \\ G_2B_2 \\ G_3B_2 \\ \vdots \\ G_7B_2 \end{matrix} \right\}$$

$P(E) = \frac{14}{27}$

↳ the event at B

Let's think about situation $\boxed{A}$:

$S = \{BB, BG, GB\}$ Homework!

!! We have to reevaluate the chances when an additional information becomes available.

- the events E_1, E_2 are independent if and only if

$$P(\underbrace{E_1 \cap E_2}_{E_1 \text{ and } E_2}) = P(E_1) \cdot P(E_2)$$

- Some mathematicians add the following formula to the axioms of probability

$$P(E_1 \cap E_2) = P(E_1) \cdot \underbrace{P(E_2 \mid E_1)}_{\text{probability of } E_2 \text{ given } E_1}$$

- define the conditional probability of B given A

$$\boxed{P(B \mid A) = \frac{P(A \cap B)}{P(A)}}$$

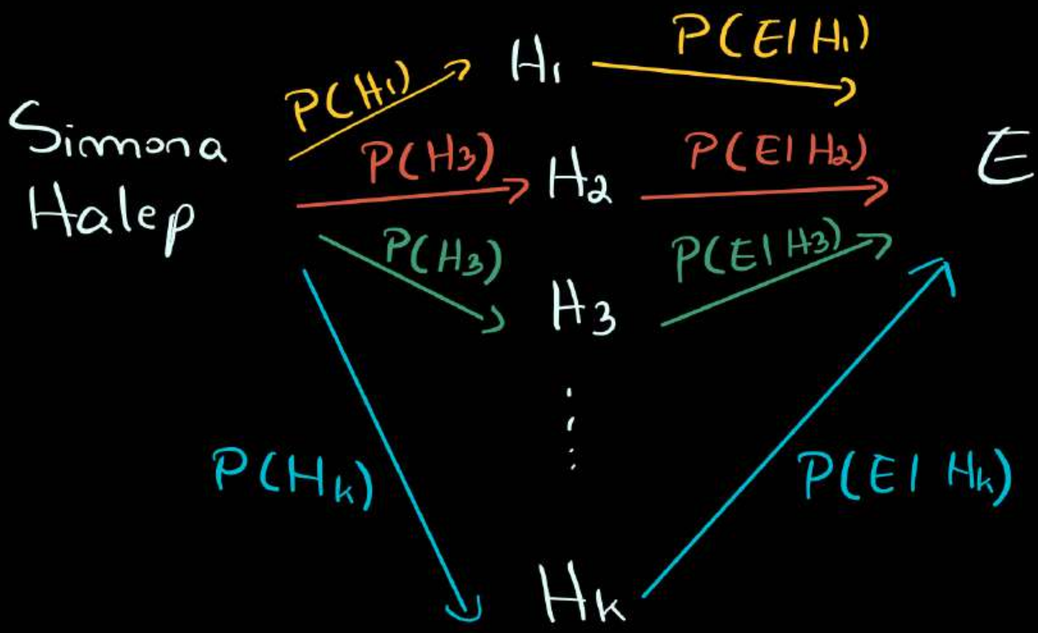
↳ probability of B having that A occurs

- Generalised:

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 \cap E_2)$$

for three events, and the general rule follows

Suppose that $H_1, H_2, \ldots H_k$ is a collection of mutually exclusive and exhaustive hypotheses, and the event E can occur in each of these hypotheses.



E is the event:

a positive drug- test

Law of total probability:

$$P(E) = P(H_1) \cdot P(E|H_1) + P(H_2) \cdot P(E|H_2) + \ldots + P(H_k) \cdot P(E|H_k)$$

Bayes law:

$$P(H_1|E) = \frac{P(H_1) \cdot P(E|H_1)}{P(E)}$$

$$P(H_i|E) = \frac{P(H_i) \cdot P(E|H_i)}{P(E)}$$

Having a deck of cards. i will slowly deal cards from a deck of 52 poker cards. You will see every card and at some point you will say "STOP".

if the next card is black => you win

if the next card is red => you lose

What's the strategy?

There is NO strategy! Ha! Gotcha :)